When the Assistant Mishears: Speech-Recognition Errors in a Voice Health-Triage Pipeline

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`https://github.com/praneeth8945/asr-error-propagation-health-triage`

Abstract

Voice assistants for health check-ins chain an automatic speech recognition (ASR) system to a text classifier and a safety layer. We measure how ASR errors propagate through such a pipeline. On a public medical speech dataset (6,661 clips, 25 symptom labels), we transcribe every clip with three Whisper models and evaluate TF-IDF, DistilBERT, keyword matching and hand-written safety rules over five leak-free splits. Whisper errors (10–23% WER) reduce DistilBERT macro-F1 from 0.79 to 0.65–0.70 and cost about 10 points of emergency recall for every method. The safety-critical keyword is frequently the misrecognised word ("heart" $\rightarrow$ "hat"). Training on Whisper transcripts gave a large gain on one split that vanished when averaged over five. Hand-written rules caught only 3–5% of emergencies. We also find that a random split inflates macro-F1 from 0.62 to 0.99 on this dataset, that Whisper WER on Indian-accented English varies 2.4-fold across speakers' first languages, and that short clips trigger hallucinated repetition.

1 Introduction

People describe symptoms in their own words, and voice interfaces promise to make health check-ins accessible to people who find forms difficult. A typical design transcribes speech with an ASR model, then routes the text through a classifier and safety rules that must catch emergencies such as chest pain or breathing difficulty. Each component is usually evaluated alone: ASR by word error rate (WER) and the classifier on clean text. This hides how the stages interact.

This study asks: *how much do Whisper transcription errors reduce symptom classification and emergency recall, and do simple fixes recover the loss?* It extends Voice Health Navigator, a local Streamlit app that uses faster-whisper, a TF-IDF topic matcher and regex safety rules. Contributions:

- An end-to-end measurement of ASR error propagation for symptom classification and emergency detection, with three Whisper sizes and five repeated leak-free splits.
- Evidence that aggregate WER understates safety impact: the emergency keyword is often the word lost.
- A negative result: training on ASR transcripts does not recover the loss on average.
- A preliminary accent analysis on Indian-accented English and a 20-sentence pilot designed to separate ASR failures from language-understanding failures.

2 Related work

Whisper [1] is trained on 680k hours of weakly supervised audio and is widely used off the shelf. ASR accuracy is known to vary by accent: Svarah [2] benchmarks English ASR on Indian accents, and AfriSpeech-200 [3] covers African-accented clinical and general speech. The medical speech dataset used here has been used for spoken symptom classification [4], and recent work benchmarks ASR coupled with language models for medical diagnostics [5]. We focus on a narrower, safety-oriented question: what happens to emergency detection when the transcript is wrong.

3 Data and method

Data. The Medical Speech, Transcription and Intent dataset (Figure Eight; Hugging Face mirror) has 6,661 short read clips, each with a human transcript and one of 25 symptom labels. Each label has only 21–35 distinct sentences, each read by many speakers. We treat *Heart hurts* and *Hard to breath* (dataset spelling) as emergency labels.

Leak-free splits. We group clips by normalised sentence text and use `StratifiedGroupKFold` (about 72/14/14 train/validation/test), so no sentence appears in two splits. All main results are averaged over five seeds.

ASR. Every clip is transcribed with Whisper tiny.en, base.en and small.en (Hugging Face `transformers`, fp16, greedy decoding). WER is computed with `jiwer` after lower-casing and removing punctuation.

Detectors. (1) TF-IDF word 1–2-grams with class-balanced logistic regression (C tuned on validation). (2) DistilBERT [6] fine-tuned for 3 epochs (lr 5×10^{-5}, batch 32), best epoch by validation macro-F1. (3) DistilBERT trained on human *and* all three Whisper transcripts of each training sentence ("Whisper-trained"). (4) Keyword matching on a fixed list (heart, chest, breath*, suffocat*, wheez*, chok*, lungs). Fuzzy and phonetic (Metaphone) variants were evaluated on validation; they added false alarms without raising recall, so plain keywords were kept. (5) The app's regex safety rules. Emergency recall counts an emergency clip as caught if it is predicted as either emergency label.

Accent and pilot. We transcribe 2,090 Svarah clips (up to 110 per first language, 19 languages) with small.en and base.en. Outputs more than 1.5$\times$ longer than the reference are flagged as hallucinations. The author also recorded 20

Transcript	WER (%)	TF-IDF	DistilBERT
Human	0.0	.68 ± .06	.79 ± .05
Whisper small.en	10.4	.60 ± .04	.70 ± .03
Whisper base.en	16.8	.58 ± .03	.67 ± .03
Whisper tiny.en	22.5	.56 ± .04	.65 ± .03

Table 1: Symptom classification macro-F1 (mean ± std, 5 splits). WER over all 6,661 clips.

Method	Human	small	base	tiny
Regex rules	.03	.05	.05	.04
TF-IDF	.76	.69	.69	.68
DistilBERT	.86	.78	.77	.77
DistilBERT (W-trained)	.88	.78	.78	.78
Keyword	**.96**	**.87**	**.86**	**.84**
Keyword + DB (W-tr.)	.96	.88	.87	.86

Table 2: Emergency recall (mean over 5 splits). Std: .04–.07 for keyword methods, .07–.21 for classifiers. False alarms 1–5%.

sentences: 10 emergencies (4 without obvious keywords) and 10 non-emergencies with sound-alike traps ("heat", "hard to sleep", "bad breath").

4 Results

4.1 A random split makes the task look solved

With a random split, 546 sentences appear in both train and test, and TF-IDF reaches 0.99 macro-F1. The leak-free split gives 0.62 for the same model. All further results use leak-free splits.

4.2 ASR errors propagate

Table 1 and Figure 1a show macro-F1 falling as WER rises. DistilBERT loses 0.09–0.14, more in absolute terms than TF-IDF (0.08–0.12): a stronger text model raises the ceiling but is not more robust to ASR errors.

4.3 Emergency detection

Table 2 shows that every method loses roughly 10 points of emergency recall on Whisper transcripts. Keyword matching is the strongest single detector (0.96 on human text, 0.84–0.87 on Whisper text, about 2% false alarms), ahead of DistilBERT (0.77–0.86) and TF-IDF (0.68–0.76). The app's regex rules caught 3% of emergencies on human transcripts: speakers say "my heart aches" rather than "chest pain".

4.4 The safety keyword is the word that breaks

When the human transcript contains "heart" (51 test clips), Whisper drops it in 12–14% of clips at every model size; tiny.en drops "breath" in 25% (5 of 20). Misrecognitions include "heart" → "hat", "harsh" and "her eggs". Even a near-correct "heartache" defeats TF-IDF, which never saw the compound in training. Aggregate WER hides this: for tiny.en, WER on emergency clips (18.9%) is lower than overall (22.5%), yet emergencies are still missed.

4.5 Training on ASR transcripts

On the first split, the Whisper-trained DistilBERT raised emergency recall on Whisper text from 0.69 to 0.86–0.90. Averaged over five splits the gain disappears (0.77–0.78 vs 0.78), and macro-F1 on human text drops from 0.79 to 0.75. The only consistent benefit is lower variance across splits. With about 9 distinct emergency sentences per test set, single-split results are unreliable.

4.6 Accent: Indian-accented English

Whisper small.en WER on Svarah is 10.2% overall and 8.5% after removing hallucinations; base.en is 27.7% and 11.1%. Across first languages, cleaned small.en WER ranges from 5.5% (Hindi) to 13.2% (Bodo) (Figure 2). Both models agree on the easiest (Hindi, Gujarati) and hardest groups (Nepali, Bodo, Assamese). Errors include regional names ("Wayanad" → "Why not?"), the same substitution of rare words for common ones seen with "heart". About 3% of clips, almost all 1–2 s long, produced runaway repetition ("Two" → "Thank you." repeated), a risk for apps that collect short spoken answers.

4.7 Pilot recordings

On the author's clear read speech, Whisper WER was 2.1–2.8% and every keyword survived. Failures moved to the detectors: keyword matching caught 6/10 emergencies (missing all four phrased without keywords, such as "I can't get enough air") and flagged "bad breath"; the regex rules caught 0/10, missing "I can't catch my breath"; DistilBERT caught 6/10 but labelled "It is hard to sleep at night" and "I have bad breath" as *Hard to breath*, suggesting reliance on surface words.

5 Discussion

The two stages fail in different ways. On crowd-recorded audio, ASR removes the words that carry the emergency meaning; on clear audio, the language-understanding stage fails on wording it has not seen. Evaluating ASR by WER and the classifier on clean text would miss both. This motivates reporting a *safety-keyword error rate* alongside WER, and evaluating triage systems end to end on spoken input. Because training on ASR text did not help, the next fixes should act on the ASR side (vocabulary biasing, confidence or N-best hypotheses) or on the representation of meaning, rather than on noise augmentation alone.

6 Limitations

Emergency test sets are small (about 9 distinct sentences per split), so recall varies by ±0.04–0.21. The emergency keywords overlap with label names, which likely flatters keyword matching. Speech is read by crowd workers, not patients, who may be distressed, in noisy settings, or code-mixing. Svarah has about 6 speakers per language and is not medical. The pilot has one speaker. Only Whisper with greedy decoding was tested.

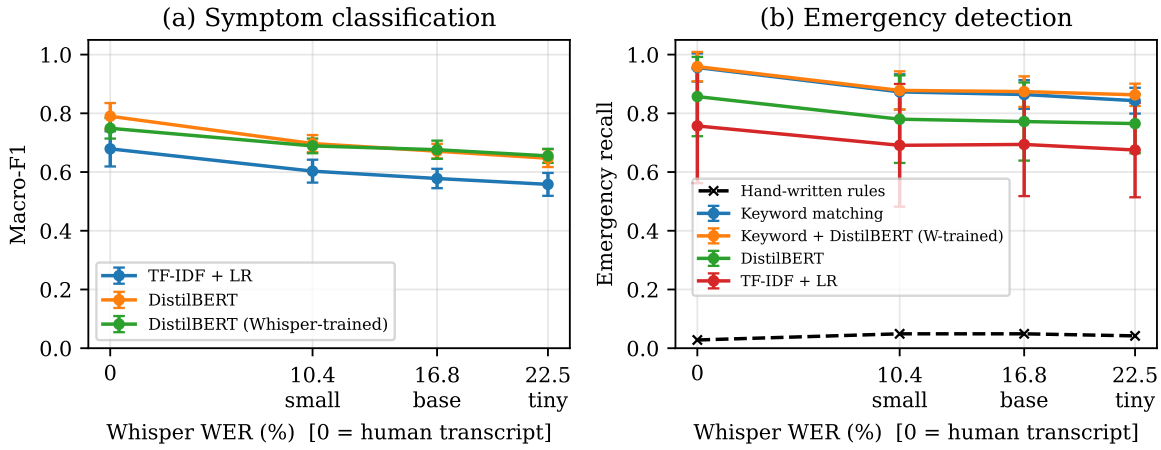


Figure 1: (a) Macro-F1 and (b) emergency recall against Whisper WER (mean ± std over 5 leak-free splits).

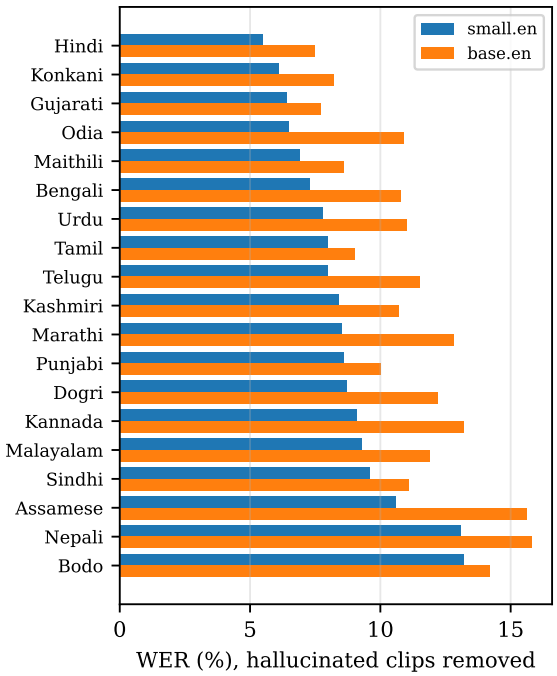


Figure 2: Whisper WER on Svarah by speaker's first language (hallucinated clips removed; about 6 speakers per language).

7 Conclusion

ASR errors cost about 10 points of emergency recall across every detector tested, concentrated in the words that matter most, and simple ASR-noise augmentation does not recover them. Repeated leak-free evaluation was essential: it exposed both a 37-point leakage inflation and a fix that did not hold up.

References

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